

Digital Signal Analysis and Processing Numerical Companion

CT 704 (BEI/BCT, IV-I) / EX 753 (BEX, IV-II) · TU, IOE · tiers over 45 papers, 2066–2082 BS

Ch 4 · Discrete Filter Structures

Numerical Problems & Solutions

42 of the 68 questions are the lattice, and they are all one recursion. Every paper is below, grouped by what the recursion is run on. Sections 3 to 5 share a single calculation; learn it once.

The step-down, which is \$3, \$4 and \$5

1. Take the polynomial $A_M(z) = 1 + a_M(1)z^{-1} + \dots + a_M(M)z^{-M}$. For an FIR $H(z) = B(z)$ it is the numerator; for an all-pole or pole-zero $H(z)$ it is **always the denominator**.
2. $k_M = a_M(M)$, read straight off the last coefficient.
- 3.

$$a_{m-1}(i) = \frac{a_m(i) - k_m a_m(m-i)}{1 - k_m^2}, \quad i = 1, \dots, m-1$$

which folds the polynomial against its reverse and drops the degree by one.

4. Repeat until A_1 , whose one coefficient is k_1 .
5. **Stable** if and only if every $|k_m| < 1$. Nothing else to compute.

Two shortcuts worth having

- The last step is always $k_1 = \frac{a_2(1)}{1 + k_2}$, because $a_2(2-1) = a_2(1)$ makes the numerator $a_2(1)(1 - k_2)$ and $1 - k_2^2 = (1 - k_2)(1 + k_2)$.
- **Keep fractions**. $1 - k^2$ breeds denominators, and a decimal answer is visibly wrong by the third stage.

The two checks

1. **Step back up:** $a_m(i) = a_{m-1}(i) + k_m a_{m-1}(m-i)$ must return the polynomial you started from. Two lines, and it catches every arithmetic slip.
2. If any $|k_m| = 1$ exactly, **stop**: the polynomial is symmetric and no lattice exists. See the box in §3.

1 Direct Form I and Direct Form II

TOP 9/45 79 Bh **2+2**, 79 Ba **4**, 82 Bh **2+2**, 74 Ch **5**, 72 Ch **5**, 72 Ka **4**, 69 Ch **4**, 68 Bh **9**, 70 Ch **4**

QUESTION AS PRINTED 2079 Bhadra · [2+2]

Obtain the Direct Form I and Direct Form II realization of the system $y[n] - 0.25y[n-2] = x[n] + 0.4x[n-1] + 0.5x[n-2]$.

Step 1: put it in computing form and read off the taps

$$y[n] = x[n] + 0.4x[n-1] + 0.5x[n-2] + 0.25y[n-2]$$

- Feedforward $b = \{1, 0.4, 0.5\}$, feedback $a = \{1, 0, -0.25\}$, so the taps are $-a_1 = 0$ and $-a_2 = +0.25$.
- $a_1 = 0$, **so there is no $y[n-1]$ branch at all**. Leave the tap out; do not draw a multiplier by zero.
- $M = N = 2$, so Direct Form I needs $2 + 2 = 4$ delays and Direct Form II needs $\max(2, 2) = 2$.

Step 2: Direct Form I

- Input into a chain of two delays, tapped 1, 0.4, 0.5 into an adder.
- That adder's output goes into a second chain of two delays, tapped 0 and 0.25 back into the same adder.
- Reading left to right it is $B(z)$ then $1/A(z)$.

Step 3: Direct Form II

- Swap the two halves and share one chain of two delays carrying $w[n]$:

$$w[n] = x[n] + 0.25w[n-2], \quad y[n] = w[n] + 0.4w[n-1] + 0.5w[n-2]$$

- Left of the chain: one branch, 0.25 from $w[n-2]$ back to the top adder. Right of the chain: three branches, 1, 0.4, 0.5 into the output adder.

- **Two delays instead of four, same arithmetic.** That is the entire answer to "why Direct Form II".

Every other paper in this group

Papers	System	Taps	Watch for
79 Ba, 82 Bh	$y[n] - 0.75y[n-1] - 0.25y[n-2]$ $= x[n] + 0.5x[n-1]$	$b = \{1, 0.5\}$ $-a = \{0.75, 0.25\}$	both feedback taps positive
74 Ch	$3y[n] + y[n-1] + 2y[n-4]$ $= 2x[n] + x[n-3]$	$b = \{\frac{2}{3}, 0, 0, \frac{1}{3}\}$ $-a = \{-\frac{1}{3}, 0, 0, -\frac{2}{3}\}$	divide by $a_0 = 3$ first ; 4 delays, most taps zero
72 Ch	$y(n) = -0.1y(n-1) + 0.2y(n-2)$ $+ 3x(n) + 3.6x(n-2) + 0.6x(n-2)$	$b = \{3, 0, 4.2\}$ $-a = \{-0.1, 0.2\}$	$x(n-2)$ is printed twice ; add them, $3.6 + 0.6 = 4.2$
72 Ka, 69 Ch	$y(n) = -0.1y(n-1) + 0.72y(n-2)$ $+ 0.7x(n) - 0.252x(n-2)$	$b = \{0.7, 0, -0.252\}$ $-a = \{-0.1, 0.72\}$	Direct Form II only
70 Ch	$y[n] - \frac{3}{4}y[n-1] + \frac{1}{8}y[n-2]$ $-x[n] - 2x[n-1] = 0$	$b = \{1, 2\}$ $-a = \{\frac{3}{4}, -\frac{1}{8}\}$	move the x terms across: they are positive on the right
68 Bh	$H(z)$ given factored, 3 zeros over 4 poles	multiply out first	see the box below

68 Bh gives $H(z)$ already factored, and wants Direct Form I and II The zeros are $\frac{1}{5}e^{\pm j\pi/5}$ and $\frac{1}{3}$; the poles are $\frac{4}{5}$, $\frac{1}{5}$ and $\frac{1}{7}e^{\pm j\pi/7}$. **A direct form needs the expanded polynomials, not the factors**, so multiply each conjugate pair out first using $(1 - re^{j\theta}z^{-1})(1 - re^{-j\theta}z^{-1}) = 1 - 2r\cos\theta z^{-1} + r^2z^{-2}$: the zero pair gives $1 - \frac{2}{5}\cos 36^\circ z^{-1} + \frac{1}{25}z^{-2}$ and the pole pair $1 - \frac{2}{7}\cos(180/7)^\circ z^{-1} + \frac{1}{49}z^{-2}$. Multiply through and you have a 3rd-order numerator over a 4th-order denominator, so Direct Form II needs **4 delays**. **The same factored form is exactly what \$2 wants left alone**, which is the joke of this chapter: cascade wants factors, direct form wants them multiplied out.

2 Cascade and parallel realizations

TOP 8/45 81 Ba **5**, 74 Ma **7**, 80 Ba **4**, 76 Ch **4**, 77 Ch **7**, 69 Bh **6**, 67 Mng, 78 Ch **6**

QUESTION AS PRINTED 2081 Baishakh · [5]

Realize the following system in cascade form of second-order sections (signal flow graph): $H(z) = \frac{10(1 - 0.25z^{-1})(1 - 0.667z^{-1})}{(1 - 0.75z^{-1})(1 - 0.125z^{-1})\{1 - (0.5 + j0.5)z^{-1}\}}$

Step 1: pair the poles, then pair the zeros to match

- **The conjugate pair must stay together.** With $\sigma = 0.5$ and $\omega = 0.5$:

$$\{1 - (0.5 + j0.5)z^{-1}\}\{1 - (0.5 - j0.5)z^{-1}\} = 1 - z^{-1} + 0.5z^{-2}$$

using $-2\sigma = -1$ and $\sigma^2 + \omega^2 = 0.5$.

- The two real poles pair with each other: $(1 - 0.75z^{-1})(1 - 0.125z^{-1}) = 1 - 0.875z^{-1} + 0.09375z^{-2}$.
- Three zeros over four poles, so one section's numerator is first order. Pair the two "small" zeros and leave $(1 + 2z^{-1})$ to ride with the real-pole section: $(1 - 0.25z^{-1})(1 - 0.667z^{-1}) = 1 - 0.917z^{-1} + 0.167z^{-2}$.

Step 2: write the two sections and place the gain

$$H(z) = 10 \cdot \underbrace{\frac{1 - 0.917z^{-1} + 0.167z^{-2}}{1 - 0.875z^{-1} + 0.09375z^{-2}}}_{\text{section 1}} \cdot \underbrace{\frac{1 + 2z^{-1}}{1 - z^{-1} + 0.5z^{-2}}}_{\text{section 2}}$$

- Draw each section in **Direct Form II** and chain them. The gain 10 goes on the input branch, or is split between sections to keep the internal signals from overflowing.
- **Every coefficient is real.** If a j survives, a pair was split.

The rest of the cascade family

All the same two steps. Only 74 Ma also wants the parallel form, which needs partial fractions instead of factors.

Papers	Poles	Sections
74 Ma	same as 81 Ba, gain 5	identical sections, gain 5; parallel form also asked , so partial-fraction the expanded $H(z)$
80 Ba	$0.5e^{\pm j\pi/3}, -0.7e^{\pm j\pi/4}$	$1 - 0.5z^{-1} + 0.25z^{-2}$ and $1 + 0.99z^{-1} + 0.49z^{-2}$
76 Ch	$0.6e^{\pm jn\pi/3}, -0.5e^{\pm j2n\pi/7}$	the paper prints a stray n in every exponent; read it as $n = 1$
77 Ch	$-0.2, 0.25, -0.5 \pm j0.15$	real pair $1 - 0.05z^{-1} - 0.05z^{-2}$, complex pair $1 + z^{-1} + 0.2725z^{-2}$
69 Bh	$0.5, 0.3, 0.7e^{\pm j\pi/4}$	$1 - 0.8z^{-1} + 0.15z^{-2}$ and $1 - 0.99z^{-1} + 0.49z^{-2}$; all-pole, numerator 1
67 Mng	$\frac{5}{6}, \frac{1}{6}, \frac{3}{4}e^{\pm j\pi/4}$	$1 - z^{-1} + \frac{5}{36}z^{-2}$ and $1 - 1.0607z^{-1} + 0.5625z^{-2}$

78 Ch asks for the parallel form from an impulse response $h[n] = (0.5)^n u[n] + n(0.2)^n u[n]$. Transform each term with the standard pairs: $\frac{1}{1 - 0.5z^{-1}}$ and $\frac{0.2z^{-1}}{(1 - 0.2z^{-1})^2}$. **Parallel form is what partial fractions already give you**, so no further work is needed: two sections added, the first a one-pole section, the second a repeated-pole section needing two delays. The repeated pole is the only wrinkle: it cannot be split further, so that section is second order with denominator $1 - 0.4z^{-1} + 0.04z^{-2}$.

67 Mng prints the same exponent twice Its last two denominator factors both read $e^{-j\pi/4}$. A real filter needs the conjugate, so the second is $e^{+j\pi/4}$, and the pair multiplies to $1 - \frac{3}{2} \cos 45^\circ z^{-1} + \frac{9}{16} z^{-2} = 1 - 1.0607z^{-1} + 0.5625z^{-2}$. Note it in one line and continue.

3 The step-down, run on an FIR numerator

TOP 9/45 81 Ba **5**, 78 Bh **3+7**, 72 Ch **5**, 73 Bh **6**, 74 Ch **5**, 74 Ash **3+7**, 73 Shr **3+7**, 76 Bh **2+5**, 72 Ka **6**, 69 Ch **6**, 70 Ch **6**

QUESTION AS PRINTED 2081 Baishakh · 2078 Bhadra · [5]

Determine the lattice coefficients corresponding to the FIR filter with system function $H(z) = A_3(z) = 1 + \frac{13}{24}z^{-1} + \frac{5}{8}z^{-2} + \frac{1}{3}z^{-3}$, and check its stability.

Stage 3

- $k_3 = a_3(3) = \frac{1}{3}$, free.
- $1 - k_3^2 = 1 - \frac{1}{9} = \frac{8}{9}$.

$$a_2(1) = \frac{a_3(1) - k_3 a_3(2)}{8/9} = \frac{\frac{13}{24} - \frac{1}{3} \cdot \frac{5}{8}}{8/9} = \frac{\frac{13}{24} - \frac{5}{24}}{8/9} = \frac{\frac{8}{24}}{8/9} = \frac{1}{3} = \frac{3}{9}$$

$$a_2(2) = \frac{a_3(2) - k_3 a_3(1)}{8/9} = \frac{\frac{5}{8} - \frac{1}{3} \cdot \frac{13}{24}}{8/9} = \frac{\frac{45}{72} - \frac{13}{72}}{8/9} = \frac{\frac{32}{72}}{8/9} = \frac{1}{2}$$

Stage 2, then stage 1

- $k_2 = a_2(2) = \frac{1}{2}$.
- Use the shortcut $k_1 = \frac{a_2(1)}{1 + k_2} = \frac{3/8}{3/2} = \frac{1}{4}$.

$$k = \left\{ \frac{1}{4}, \frac{1}{2}, \frac{1}{3} \right\}$$

- **Stable:** every $|k_m| < 1$.
- Check by stepping back up: $a_2(1) = k_1 + k_2 k_1 = \frac{1}{4} + \frac{1}{8} = \frac{3}{8} \checkmark$, and $a_3(1) = a_2(1) + k_3 a_2(2) = \frac{3}{8} + \frac{1}{3} \cdot \frac{1}{2} = \frac{13}{24} \checkmark$.

74 Ch is the identical question with the coefficients written as $\frac{52}{96}$ and $\frac{25}{40}$, which reduce to $\frac{13}{24}$ and $\frac{5}{8}$. Same answer, $k = \{\frac{1}{4}, \frac{1}{2}, \frac{1}{3}\}$. So is 79 Bh, backwards: see §4.

The other FIR papers

Papers	$H(z)$	$k = \{k_1, k_2, k_3, \dots\}$	Stable?
81 Ba, 78 Bh, 74 Ch	$1 + \frac{13}{24}z^{-1} + \frac{5}{8}z^{-2} + \frac{1}{3}z^{-3}$	$\{\frac{1}{4}, \frac{1}{2}, \frac{1}{3}\}$	yes
73 Shr	$2 + 1.8z^{-1} - 1.6z^{-2} + z^{-3}$ (divide by 2 first)	$\{-2.6, -1.6667, 0.5\}$	no
76 Bh	$1 + 2z^{-1} + 0.62z^{-2} + 0.8z^{-3}$	$\{-2.4258, -2.7222, 0.8\}$	no
72 Ka, 69 Ch	$1 + 2z^{-1} - 3z^{-2} + 4z^{-3}$	$\{-0.5385, 0.7333, 4\}$	no
70 Ch	$1 + 3.1z^{-1} + 5.5z^{-2}$ $+ 4.2z^{-3} + 2.3z^{-4}$	$\{0.3381, 1.1664, 0.6830, 2.3\}$	no

Five papers ask for a lattice that does not exist 72 Ch $(1 + 2z^{-1} + z^{-2})$, 73 Bh $(1 + 2z^{-1} + 2z^{-2} + z^{-3})$, 74 Ash $(1 + 0.7z^{-1} + 1.2z^{-2} - z^{-3})$, 75 Ch and 73 Ch (both containing $1 + \frac{2}{3}z^{-1} + \frac{5}{8}z^{-2} + \frac{2}{3}z^{-3} + z^{-4}$). In each the last coefficient has magnitude 1, so $k_M = \pm 1$ and the step-down must divide by $1 - k_M^2 = 0$. **Four of the five are symmetric polynomials**, that is linear-phase filters, and symmetry forces $a_M(M) = a_M(0) = 1$. **A linear-phase FIR filter has no lattice realization**, and that is the answer: show $k_M = 1$, show the division by zero, and note that the zeros lie on the unit circle. For 72 Ch it is visible without any recursion, since $1 + 2z^{-1} + z^{-2} = (1 + z^{-1})^2$ is a double zero at $z = -1$.

4 The reverse: lattice coefficients to direct form

PIN 4/45 79 Bh 6, 71 Shr 5, 80 Ch, 74 Ma 7

QUESTION AS PRINTED 2079 Bhadra · [6]

Given a 3-stage lattice filter with coefficients $K_1 = \frac{1}{4}$, $K_2 = \frac{1}{2}$, $K_3 = \frac{1}{3}$, obtain the system function and the FIR filter coefficients.

- Step up, one stage at a time, with $a_m(i) = a_{m-1}(i) + k_m a_{m-1}(m-i)$ and $a_m(m) = k_m$.
- $A_1(z) = 1 + \frac{1}{4}z^{-1}$.
- $a_2(1) = a_1(1) + k_2 a_1(1) = \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{4} = \frac{3}{8}$, and $a_2(2) = k_2 = \frac{1}{2}$, so $A_2(z) = 1 + \frac{3}{8}z^{-1} + \frac{1}{2}z^{-2}$.
- $a_3(1) = a_2(1) + k_3 a_2(2) = \frac{3}{8} + \frac{1}{3} \cdot \frac{1}{2} = \frac{13}{24}$; $a_3(2) = a_2(2) + k_3 a_2(1) = \frac{1}{2} + \frac{1}{3} \cdot \frac{3}{8} = \frac{5}{8}$; $a_3(3) = k_3 = \frac{1}{3}$.

$$H(z) = A_3(z) = 1 + \frac{13}{24}z^{-1} + \frac{5}{8}z^{-2} + \frac{1}{3}z^{-3}$$

- So $h[n] = \{1, \frac{13}{24}, \frac{5}{8}, \frac{1}{3}\}$.
- **This is §3 run backwards**, on the same numbers. 71 Shr asks it identically.

80 Ch and 74 Ma change one coefficient They give $K_1 = \frac{1}{4}$, $K_2 = \frac{1}{4}$, $K_3 = \frac{1}{3}$. The same recursion gives $a_2(1) = \frac{1}{4} + \frac{1}{4} \cdot \frac{1}{4} = \frac{5}{16}$, $a_2(2) = \frac{1}{4}$, then $a_3(1) = \frac{5}{16} + \frac{1}{3} \cdot \frac{1}{4} = \frac{19}{48}$, $a_3(2) = \frac{1}{4} + \frac{1}{3} \cdot \frac{5}{16} = \frac{17}{48}$, $a_3(3) = \frac{1}{3}$. So $h[n] = \{1, \frac{19}{48}, \frac{17}{48}, \frac{1}{3}\}$. **The two thirds in the answer look alike but are not:** $\frac{19}{48}$ and $\frac{17}{48}$ differ, and writing both as $\frac{18}{48} = \frac{3}{8}$ is the easy mistake.

5 All-pole and lattice-ladder

TOP 21/45 80 Bh 10, 81 Ch, 72 Ash 4+2+1, 79 Ch 4+2+1, 69 Bh 8, 71 Bh 6+2, 70 Ma 6, 71 Ch 4+2+1, 80 Ba, 82 Bh 6, 82 Ba 10, 79 Ba 6+1, 81 Bh 6+4, 76 Ch 6+4, 76 Ash 6+3, 74 Bh 5+1, 75 Bh 7+1, 77 Ch 7+2, 73 Ma 6+1, 78 Ch 6+2, 70 Asa 6, 66 Ma 6, 70 Bh 8

QUESTION AS PRINTED 2080 Bhadra · [10]

Determine the lattice coefficients and draw the lattice structure for the system $H(z) = \frac{1}{1 - 0.2z^{-1} + 0.4z^{-2} + 0.6z^{-3}}$.

The denominator is the polynomial: everything else is identical to §3

- $A_3(z) = 1 - 0.2z^{-1} + 0.4z^{-2} + 0.6z^{-3}$, so $k_3 = 0.6$ and $1 - k_3^2 = 0.64$.

$$a_2(1) = \frac{-0.2 - 0.6(0.4)}{0.64} = \frac{-0.44}{0.64} = -\frac{11}{16}, \quad a_2(2) = \frac{0.4 - 0.6(-0.2)}{0.64} = \frac{0.52}{0.64} = \frac{13}{16}$$

- $k_2 = a_2(2) = \frac{13}{16} = 0.8125$.
- $k_1 = \frac{a_2(1)}{1 + k_2} = \frac{-11/16}{29/16} = -\frac{11}{29} = -0.3793$.

$$k = \left\{ -\frac{11}{29}, \frac{13}{16}, \frac{3}{5} \right\}$$

- All $|k_m| < 1$, so the system is **stable**. **An all-pole $H(z)$ is drawn as the lattice run in reverse**: the same stages, with the forward path fed from the output end.

Adding the ladder, when the numerator is not 1

- Run the step-down on $A(z)$ exactly as above and **keep every intermediate** $A_i(z)$.
- Then work down from $C_M = b_M$ with $C_m = b_m - \sum_{i=m+1}^M C_i a_i(i-m)$.
- **The k 's never depend on the numerator**, so papers sharing a denominator share their lattice: 76 Ch with 76 Ash, and 74 Bh with 75 Bh.

Papers	$H(z)$ denominator	k	C (ladder)
80 Bh	$1 - 0.2z^{-1} + 0.4z^{-2} + 0.6z^{-3}$	$\{-0.3793, 0.8125, 0.6\}$	all-pole
81 Ch, 72 Ash	$1 - 0.525z^{-1} + 0.6125z^{-2} + 0.3z^{-3}$	$\{-0.4219, 0.8462, 0.3\}$	all-pole, stable
79 Ch, 69 Bh	$1 - 0.9z^{-1} + 0.64z^{-2} - 0.576z^{-3}$	$\{-0.6728, 0.1820, -0.576\}$	all-pole, stable
71 Bh	$1 - 0.3z^{-1} + 0.5z^{-2} + 0.25z^{-3}$	$\{-0.2810, 0.6133, 0.25\}$	all-pole
70 Ma	$1 + 2z^{-1} - 3z^{-2} + 4z^{-3}$	$\{-0.5385, 0.7333, 4\}$	unstable , same as 72 Ka
71 Ch	$1 - 0.01z^{-1} - 0.23z^{-2} + 0.5z^{-3}$	$\{0.2, -0.3, 0.5\}$	all-pole, stable
80 Ba, 82 Bh	$1 - 0.3z^{-1} + 0.25z^{-2}$, $B = 2 - 0.7z^{-1} + 0.5z^{-2}$	$\{-0.24, 0.25\}$	$\{1.743, -0.55, 0.5\}$
79 Ba	$1 - 0.3z^{-1} + 0.5z^{-2}$, $B = 1 - 0.4z^{-1} + 0.25z^{-2}$	$\{-0.2, 0.5\}$	$\{0.81, -0.325, 0.25\}$
76 Ch	$1 - 0.5z^{-1} - 0.7z^{-2} + 0.3z^{-3}$, $B = 0.5 - 2z^{-1} + 3z^{-2}$	$\{-0.8056, -0.6044, 0.3\}$	$\{1.4722, -1.0440, 3\}$
76 Ash	same denominator , $B = 0.7 - 1.5z^{-1} + 0.5z^{-2}$	$\{-0.8056, -0.6044, 0.3\}$	$\{-0.0778, -1.3407, 0.5\}$
74 Bh	$1 + 0.27z^{-1} + 0.06z^{-2} - 0.75z^{-3}$, $B = 0.62 + 0.42z^{-1} - 0.25z^{-2}$	$\{0.45, 0.6, -0.75\}$	$\{0.5, 0.6, -0.25\}$
75 Bh	same denominator , $B = 0.6 - 0.45z^{-1} - 0.25z^{-2}$	$\{0.45, 0.6, -0.75\}$	$\{0.8715, -0.27, -0.25\}$
77 Ch	$1 + 0.2z^{-1} + 0.7z^{-2} - 0.75z^{-3}$, $B = 0.5 - 0.4z^{-1} - 0.2z^{-2}$	$\{0.5631, \mathbf{1.9429}, -0.75\}$	unstable , $ k_2 > 1$
73 Ma	$1 - 0.45z^{-1} + 0.3z^{-2} + 0.5z^{-3}$, $B = 0.5 + 0.65z^{-1} + 0.2z^{-2}$	$\{-0.4706, 0.7, 0.5\}$	$\{0.7412, 0.81, 0.2\}$
78 Ch	$1 + 0.2z^{-1} - 0.15z^{-2}$, $B = 1 - z^{-1} + 0.5z^{-2}$	$\{0.2353, -0.15\}$	$\{1.3338, -1.1, 0.5\}$
70 Asa	$1 - 0.3z^{-1} + 0.4z^{-2}$, $B = 0.5 + 2z^{-1} + 0.6z^{-2}$	$\{-0.2143, 0.4\}$	$\{0.7271, 2.18, 0.6\}$
66 Ma	$1 + \frac{1}{5}z^{-1} + \frac{2}{5}z^{-2} + \frac{3}{5}z^{-3}$, $B = \frac{1}{2} + \frac{1}{3}z^{-1} + \frac{1}{4}z^{-2} + \frac{1}{5}z^{-3}$	$\{-0.0435, 0.4375, 0.6\}$	$\{0.2997, 0.2665, 0.21, 0.2\}$
82 Ba	$1 - 0.001z^{-1} + 0.6546z^{-2} - 0.0775z^{-3}$	$\{0.0302, 0.6585, -0.0775\}$	$\{-0.0493, 0.3059, 0.5124, 0.2759\}$
81 Bh	$(1 + 0.5z^{-1})(1 + 0.3z^{-1})(1 + 0.4z^{-1})$ $= 1 + 1.2z^{-1} + 0.47z^{-2} + 0.06z^{-3}$	multiply out first	$B = 1 + z^{-1} + z^{-2}$

70 Bh: the all-pass, and the tidiest result in the chapter $H(z) = \frac{1 + 0.4z^{-1} - 1.2z^{-2} + 2z^{-3}}{2 - 1.2z^{-1} + 0.4z^{-2} + z^{-3}}$. Look before calculating: **the numerator is the denominator written backwards**. That is the definition of an **all-pass**, $H(z) = z^{-N}A(z^{-1})/A(z)$. The backward path of a lattice already computes $z^{-N}A(z^{-1})$, so the ladder collapses completely: every $C_m = 0$ except $C_3 = 1$. **An all-pass is a plain lattice with the output taken from the last backward node**. Normalising by $a_0 = 2$ gives $A(z) = 1 - 0.6z^{-1} + 0.2z^{-2} + 0.5z^{-3}$ and $k = \{-0.56, \frac{2}{3}, \frac{1}{2}\}$, all inside the unit circle, so it is stable.

6 Quantization, number formats and limit cycles

TOP 10/45 71 Ch **3**, 70 Asa **1+3**, 72 Ma **5**, 81 Ch, 80 Ch, 70 Bh **3+4**, 66 Ma **4**, 71 Shr **1+1+2+1**, 69 Bh **6**, 69 Bh **3+4**, 75 Ash **3**, 67 Mng

QUESTION AS PRINTED 2066 Magh · [4]

For the first order filter $y(n) = Q\{a y(n-1)\} + x(n)$, the product term $a y(n-1)$ has been quantized by rounding it to 3 bits, with $y(-1) = 0$, $x(n) = 0.875\delta(n)$ and $a = -0.5$. Show whether the filter goes into a limit cycle. What is the period?

Just run it. Three bits after the point means the step is $q = 2^{-3} = \frac{1}{8}$

n	$a y(n-1)$	$Q\{\cdot\}$ to $\frac{1}{8}$	$y(n)$	
0	$-0.5 \times 0 = 0$	0	0.875	the impulse arrives
1	-0.4375	-0.5	-0.5	rounds away from zero
2	0.25	0.25	0.25	exact, no rounding needed
3	-0.125	-0.125	-0.125	exact; now at the dead-band edge
4	0.0625	0.125	0.125	half a step, rounded up
5	-0.0625	-0.125	-0.125	and back again
6	0.0625	0.125	0.125	

– **Yes, it enters a limit cycle**, amplitude $\frac{1}{8}$, and it never leaves.

– **Period 2**, because $a < 0$ flips the sign each step.

– Confirm with the dead band: $\frac{q}{2(1-|a|)} = \frac{1/8}{2(1-0.5)} = \frac{1}{8}$, **exactly the amplitude observed**. Quote the formula and the table together.

- The mechanism is row 4 alone: $-0.5 \times (-0.125) = 0.0625$ is *half* a step, and rounding lifts it back to a whole one instead of letting it fall to zero.

75 Ash: $\pm \frac{5}{8}$ in the three fixed-point codes

- $\frac{5}{8} = 0.101_2$, so with a sign bit and 3 fraction bits:
 $+\frac{5}{8} = 0.101$ in sign magnitude, 1's complement and 2's complement alike.
 $-\frac{5}{8}$: sign magnitude 1.101 (flip the sign bit only), 1's complement 1.010 (invert the fraction bits), 2's complement 1.011 (invert and add 1 in the last place).
- **Check: 1's complement plus 1 ulp is 2's complement.** $1.010 + 0.001 = 1.011 \checkmark$.

69 Bh asks for $-192/220$ in 8-bit form, which cannot be done $192/220 = 0.87273$, and 220 is not a power of two, so **the number has no exact binary fraction** in 8 bits or any other number of bits. Rounded to 8 bits it is $-\frac{223}{256} = -0.871094$, giving sign magnitude 1.11011111, 1's complement 1.00100000 and 2's complement 1.00100001. **Almost certainly 220 is a misprint for 256**, the natural 8-bit scale: $-192/256 = -0.75$ exactly, sign magnitude 1.11000000, 1's complement 1.00111111, 2's complement 1.01000000. Give the clean version and note the other in a line.

67 Mng: show the two truncation error ranges by example Take $b_u = 6$ bits truncated to $b = 3$, so $q = \frac{1}{8}$ and $q_u = \frac{1}{64}$, and $q - q_u = \frac{7}{64}$. Sweeping every 6-bit fraction confirms both bounds exactly. **Sign magnitude:** truncating shortens the *magnitude*, so a positive number falls and a negative number rises, and e runs from $-\frac{7}{64}$ to $+\frac{7}{64}$, **symmetric**. **2's complement:** truncating always drops the value towards $-\infty$ whatever the sign, so e runs from $-\frac{7}{64}$ to 0 and **never exceeds zero**. One example each way is enough: $+0.828125 \rightarrow 0.75$ (error $-\frac{7}{64}$ in both codes) and $-0.828125 \rightarrow -0.75$ in sign magnitude (error $+\frac{7}{64}$) but $\rightarrow -0.875$ in 2's complement (error $-\frac{3}{64}$). **That asymmetry is why 2's complement truncation adds a DC bias** and rounding is preferred.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full. $\rightarrow$ reads "leads to / therefore", $+$ reads "and", $=$ reads "is". $\checkmark$ means the property holds, $\times$ that it does not.

Short	Means	Short	Means
DSP	digital signal processing	DSAP	digital signal analysis and processing
DT	discrete-time	CT	continuous-time
LTI	linear time-invariant	LSI	linear shift-invariant, the same thing
TI	time-invariant	BIBO	bounded input, bounded output
LCCDE	linear constant-coefficient difference equation	ROC	region of convergence
DTFS	discrete-time Fourier series	DTFT	discrete-time Fourier transform
CTFT	continuous-time Fourier transform	DFS	discrete Fourier series
DFT	discrete Fourier transform	IDFT	inverse DFT
FFT	fast Fourier transform	DIT	decimation in time
DIF	decimation in frequency	MAC	multiply and accumulate
FIR	finite impulse response	IIR	infinite impulse response
LPF	low pass filter	HPF	high pass filter
BPF	band pass filter	BSF	band stop filter
BLT	bilinear transformation	II	impulse invariance
A/D	analog to digital	D/A	digital to analog
S/H	sample and hold	ZOH	zero-order hold
SNR	signal to noise ratio	PYQ	past-year question
coeff	coefficient	freq	frequency
eqn	equation	no.	number
ops	operations	attr	attribute
w/	with	w/o	without
Eg	for example	Vs	compare / differentiate
+Fig	a figure is wanted	(I)	importance / need
(+)	advantage	(-)	disadvantage

Exam paper codes

Month codes

- **Ba** Baishakh **Jth** Jestha **Asa** Ashad
- **Shr** Shrawan **Bh** Bhadra **Ash** Ashwin
- **Ka** Kartik **Mng** Mangsir **Po** Poush
- **Ma** Magh **Ch** Chaitra
- **Ash and Asa are different months:** 2075 Ashwin and 2070 Ashad are separate papers.

How a year code is written

- **Bold** = **Regular** exam. Plain = Back exam.
 - “New Back (2066 & Later Batch)” counts as Back.
 - A header printing “Regular / Back” in one box counts as Regular.
- Typewriter = the EX 753 paper (DSP, BEX, IV/II).
Normal font = **CT 704** (DSAP, BEI/BCT, IV/I).
- The two axes are independent: **76 Ch** is CT 704 Regular, 76 Ash is CT 704 Back, 76 Bh is EX 753 Regular, 74 Ma is EX 753 Back.
- **69 Bh and 69 Bh are two different papers**, one in each course.

Frequency chips

- **TOP n/45** asked in **10 or more** of the 45 papers.
- **HOT n/45** asked in **4 to 9** papers.
- **PIN n/45** asked in **1 to 3** papers.
- **verified/added** written or corrected against a reference because the source notes did not cover it, or covered it wrongly.
- **5** marks the question has actually carried. Several chips mean the same question has been set at different marks in different years.
- The count is over **45 papers**: 28 CT 704 (2066 Magh to 2082 Bhadra) and 17 EX 753 (2069 Bhadra to 2081 Chaitra).
- Counts come from the Detailed PYQ document, which reproduces every question in full.